



Lok Jagruti Kendra University
Ahmedabad – 382210

**Syllabus for Five Years Master of Science (Information
Technology) Integrated Programme
Semester 4**

(Effective from Academic 2021)

Course Code	050020401			
Category	Core Subject			
Course Title	Object Oriented Programming using Java			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	4	0	4	6
Pre-requisites (if any)	Basic knowledge of programming in JAVA			

1. Course Objectives:

- To understand how a problem is solved step by step to get the desired output.

2. Course Contents

Unit	Course Content	Weightage
I	Introduction to Java Introduction: Introducing Object-Oriented Programming Concepts, The Origins of Java, Features of Java, Java's magic: The Bytecode, Obtaining the java development kit, understanding the class and main () method, First simple program, compile and execute the program, Comments, The Java Keywords, Identifiers in Java, The Java Class Libraries. Data Types and Operators : Primitive Data Types – Boolean, Numeric, Character, Literals Hexadecimal, Octal, and Binary Literals, Character Escape Sequences, String Literals, A Closer Look at Variables - Initializing a Variable, Dynamic Initialization, The Scope and Lifetime of Variables, Operators – Arithmetic, Relational, Logical, Bitwise, Increment – Decrement, Conditional Operators, The Assignment Operator, Shorthand Assignments, Type Conversion in Assignments, Casting Incompatible Types, Operator Precedence. Program Control Statements : The if Statement, Nested ifs, The if-else-if Ladder, The switch Statement, Nested switch Statements, The for Loop Some Variations on the for Loop, Missing Pieces, The Infinite Loop, Loops with No Body, For Each loop, The while Loop, The do-while Loop, Nested Loops, break, continue, goto and return statements	20%
II	OOP Fundamentals Classes, Objects and Methods: Class Fundamentals, The General Form of a Class, Defining a Class, How Objects Are Created, Method, Returning from a Method, Constructors [Default, Parameterized, Copy Constructors], This keyword. A Closer Look at Methods and Classes: Pass Objects to Methods, Returning objects, Method Overloading, Constructor Overloading,	30%

	Understanding Static variable, Static method and Static Block, Constants, Introducing Nested and Inner Classes, Inheritance : Inheritance Basics, Types of Inheritance, Member Access and Inheritance, Constructors and Inheritance, Using super to Call Superclass Constructors, Using super to Access Superclass Members, Creating a Multilevel Hierarchy, When Are Constructors Executed?, Superclass References and Subclass Objects, Method Overriding, Overridden Methods Support Polymorphism, Using final - final Prevents Overriding, final Prevents Inheritance, Using final with Data Members, Learning about Information Hiding	
III	Advance Java Concepts Abstract Class :Introduction, Abstract Class Interfaces: Interface, Implementing Interfaces, Using Interface References, Variables in Interfaces, Interfaces Can Be Extended, Default Interface Methods (Default Method Fundamentals, Use static Methods in an Interface, Private Interface Methods Packages: Packages (Defining a Package, Finding Packages and CLASSPATH, A Short Package Example), Packages and Member Access (A Package Access Example), Understanding Protected Members, Importing Packages, Java's Class Library Is Contained in Packages, Exception Handling: The Exception Hierarchy, Exception Handling Fundamentals, Using try and catch, A Simple Exception Example, Using Multiple catch Statements, Catching Subclass Exceptions, Try Blocks Can Be Nested, Throwing an Exception, Rethrowing an Exception, A Closer Look at Throwable, Using finally, Using throws, Java's Built-in Exceptions, User-Defined Exceptions	30%
IV	Characters, Strings and String Buffer: Manipulating Characters, Declaring a String Object, Comparing String Values, Using other String Methods, Converting Strings to Numbers, StringBuffer Class, Methods of StringBuffer Class Wrapper Classes (Autoboxing) : Introduction, The Number class, The Byte Class, The Short Class, The Integer Class, The Long Class, The Float Class, The Double Class, The Boolean Class Using I/O: Java's I/O Is Built upon Streams, Byte Streams and Character Streams, The Byte Stream Classes, The Character Stream Classes, The Predefined Streams, Using the Byte Streams (Reading Console Input, Writing Console Output), Reading and Writing Files Using Byte Streams (Inputting from a File, Writing to a File), Automatically Closing a File, Reading and Writing Binary Data	20%
V	Accessing Databases with JDBC, Loading JDBC Driver, Establishing Connection, Creating Statements, Executing SQL and Processing Results of a Query, Using Prepared Statement, Using CallableStatement, Using Database Transactions	Only for Practicals

Desirable:

- Seminar/ workshop on topics :
 - **Arrays:** One-Dimensional Arrays, Two-Dimensional Arrays, Declaring and Initializing an Array, Passing Arrays to a Methods, Using the Arrays Class, Iterating the array using for each loop.
 - **Varargs:** Variable-Length Arguments , Varargs Basics, Overloading Varargs Methods, Varargs and Ambiguity

3. Text Books:

- 1) Herbert Schildt, Java™: A Beginner's Guide, 7th Edition
- 2) Programming with Java A Primer – E.Balaguruswamy, McGrawhill
- 3) Hari Mohan Pandey, JAVA Programming, 978-8131733110, Pearson
- 4) Paul J. Deitel, Harvey M. Deitel, Java SE8 for Programmers, ISBN: 9789332539068, Pearson
- 5) Cay S Horstmann, Gary Cornell, “Core Java 2, Volume 1 – Fundamentals”, Pearson Education (8th edition – 2008).
- 6) Ivor Horton's “Beginning Java 2” JDK 5 Edition, Wiley Computer Publishing, (2007).

4. Webilography :

- 1) <https://www.javatpoint.com/>

5. Accomplishment of the student after completing the course: :

- Ability to design and implement various abstract data types.
- Understand the applications of various data structures in the basic computer system.
- Understand and implement the various sorting and Searching methods on the data sets(big and small).

Course Code	050020402			
Category	Core Subject			
Course Title	Database Management Systems II (PL/SQL)			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	2	1	4	5
Pre-requisites (if any)	Basic knowledge of SQL			

1. Course Objectives:

- To understand the fundamental concepts of Transaction Management.
- To understand the fundamental concepts of Concurrency Control
- To be able to understand various database objects (Stored procedure, function and Packages)
- To be able to understand about various database triggers
- To design domain constraints, web services
- To understand necessary concept for web service implementation

2.CourseContents:

Unit	Course Content	Weightage
Unit I	Advanced SQL : Top-n Query, Correlated Queries, Sub query Oracle Database Overview Link Server (MS SQL) Overview Relational Database design by ER and EER to Relational Mapping, Mapping EER model construct to Relations	20%
Unit II	Transaction Management : - Transaction Concepts, properties, states, implementations of Atomicity and Durability, Concurrent Executions, Serializability, and Recoverability. Concurrency Control :- Concurrency Control- Overview, Lock based protocol, Timestamp based protocol, Concurrency control problems, concurrency control with time stamping methods	20%
Unit III	Basics of PL/SQL Overview, PL/SQL, Advantages of PL/SQL, Generic PL/SQL Block PL/SQL Fundamentals: PL/SQL Variables and PL/SQL Data types, Variable attribute (%type,%rowtype) PL/SQL Control Structure (looping)	20%
Unit IV	PLSQL Advanced PL/SQL Cursor Management : Types of cursor (Implicit & Explicit) , Cursor Management, Parameterized cursor, for loop cursor Error handling in PL/SQL & Security : Predefined, user defined Error handling, SQLCODE,SQLERRM,Raise_application_error	20%

Unit V	PL/SQL Database Objects and Triggers Sub program: Stored procedures & Functions, Packages, Triggers: Overview Types of triggers, Application of Triggers	20%
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Desirable:

- Developing a Web Service from a PL/SQL Package
- Seminar on Object Types, Nested Table, Cluster

3.Text Books:

1. Ivan Byross, SQL, PL/SQL the programming Language of Oracle, BPB
2. Kevin Loney, Oracle Database 11g :The Complete Reference, Oracle Press, McGrawHill
3. Oracle9i PL/SQL Programming Scott Urman Oracle Press – TMH
4. Alexis Leon, Mathews Leon, “Essentials of Database Management Systems” (Second reprint 2009), Tata McGraw Hill Publication
5. S. K. Singh, “Database Systems: Concepts, Design and Applications”, Pearson Education
6. RamezElmsari, Shamkant B Navathe, “Fundamentals of Database Systems”, Pearson Education, 7th Edition

4 Webilography:

1. <https://docs.oracle.com/en/database/index.html>
2. <https://docs.oracle.com/database/121/SQLRF/toc.htm>

5 Accomplishment of the student after completing the course:

- A student would be able to effectively squeeze the “real world” data into the relational data model of the database system and data retrieval
- Clear understanding for the need of procedure, function and package.
- Understand the uses the triggers
- Understand the need for back end programming

Course Code	050020403			
Category	Interdisciplinary			
Course Title	Statistical Analysis using R programming			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	3	0	2	4
Pre-requisites (if any)	-			

1. Course Objectives:

- To present the general theory of statistical distributions as well as the standard distributions found in statistical practice.
- To learn scientific view to conduct the survey in proper way to collect the data about specific perspective.
- To Learn variety of probability and nonprobability sampling methods for selecting a sample from a population
- To learn how to develop regression model and apply for the specific perspective data appropriate manner.

2. Course Contents:

Module	Content	Weightage
Unit I	Introduction to Statistics and Descriptive Statistics Introduction: Broad areas (classification) of Statistics, Applications of Statistics in Computer science Data: Data and information, Types of Data : Qualitative (Nominal, Ordinal) and Quantitative (Discrete, continuous) Descriptive Statistics: Central Tendency, Mean and its Characteristics, Median and its Characteristics, Mode and its Characteristics, Range, Quartiles (Q1, Q2, Q3), Inter quartile Range (IQR), Percentile. Describing Data Visually: Frequency Distributions and Histograms, Pie Charts, Bar Charts, Pareto Chart, Scatter Plots (Degree of Association); Line Charts, Box Plots	15%
Unit II	Measures of Dispersion, Skewness, Measures of association and regression analysis Dispersion: Variance, Standard Deviation and its Characteristics, Coefficient of Variation Skewness: Definition and its types, Measures of Coefficient of Skewness Measures of Association: Covariance, Correlation, Coefficient of Correlation	15%

	Regression: Introduction, Independent and dependent data, Simple Regression Analysis, Least Square method, Fitting of regression line (Y On X and X on Y), Prediction of dependent data using regression line	
Unit III	Distribution, Distribution functions and Measure of Central Tendency Introduction: Common Framework: Experiment, Event, Elementary Events, Sample Space; Definition of Probability; Marginal Probability; Probability of Union of Events (Addition Laws), Probability Matrix; Probability of Complement of a Union; Probability of Joint Events (General Laws of Multiplication); Conditional Probability; Mutually Exclusive Events, Independent Events Discrete Probability Distributions: Introduction, Binomial Distribution, Poisson Distribution, Applications Continuous Probability Distributions: Introduction, Normal Distribution, Exponential Distribution, Applications	25%
Unit IV	Sampling, Sampling Distributions and Estimation Types of Sampling: Random, Non-random; Sampling Distribution of $\bar{x}$; Central Limit Theorem; z Formula for Sample Mean; Standard Error of Mean; Sampling from a Finite Population Estimation for Single Population: Estimating the Population Mean using z Statistic (σ Known); Estimating the Population Mean using the z Statistic when the Sample Size is Small; Estimating the Population Mean using t Statistic (σ Unknown)	25%
Unit V	Hypothesis Testing and Analysis of variance Hypothesis Testing : Null Hypothesis, Alternate Hypothesis; Type I & Type II Errors, Testing Hypotheses about a Population Mean using z Statistic (σ Known); Using Critical Value Method to test Hypotheses, Examples; Population Mean Testing Hypotheses about a Population Mean using t Statistic (σ Unknown); Testing Hypotheses about a Proportion Analysis of variance: One way analysis of Variance	20%

Note: Proofs of Theorems not required

3. Text Books:

1. Ken Black, “Business Statistics for Contemporary Decision Making”, Wiley Student Edition, 2010
2. David P. Doane, Lori E. Seward, “Applied Statistics in Business and Economics” Tata McGraw-Hill, 2010
3. Anderson, Sweeney, Williams, “Statistics for business and economics”, 9th edition, Thompson Publication
4. Bharat Jhunjhunwala, “Business Statistics”, first edition, S Chand, 2008
5. Richard Levin, David Rubin, “Statistics for Management”, 7th edition, PHI
6. Nabendu Pal, Sahadeb Sarkar, “Statistics-Concepts and Applications”, 2nd edition, PHI
7. J. Susan Milton & Jesse Arnold, “Introduction to Probability & Statistics: Principles & Applications for Engineering & Computing Sciences”, McGraw-Hill Education
8. S P Gupta, “Statistical Methods”, 30th edition, S Chand

4. Bibliography**5. Accomplishment of the student after completing the course:**

- This course will enhance the student’s ability to think logically and mathematically.

Course Code	050020404			
Category	Core Subject			
Course Title	Operating Systems and Virtualization			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	3	1	2	5
Pre-requisites (if any)	-			

1. COMPETENCY (Programme Outcome (PO) according to NBA Terminology):

- To install & configure various Operating Systems and basic shell scripts.

2. Course Contents :

Module	Major Outcomes (Course Outcomes in Cognitive Domain according to NBA terminology)	Learning (Course in Domain to NBA terminology)	Content	Weightage
I	Explain operating system	operating	Operating System Concepts Operating System Objectives and Functions, Evolution of Operating System, Needs of an Operating System	10%
	Explain types of operating system	types of operating system	Types of OS : Batch, Multi-Programming, Time sharing, Real-Time, Multitasking, Multi-threading Operating System Structure: Layered System, Microkernel and Virtual Machine Case Study: Linux, Windows	
II	Describe process, process state and process model	process, process state and process model	Processor and Process Management Process Description and Control: Process States, Process Description, Process Control, UNIX Process Management.	25%
	Differentiate ULT and KLT.	ULT and KLT.	Thread : Thread, (uses, applications, Types:(User Level Thread and Kernel Level Thread) and Multithreading),	

	Compare processor scheduling algorithm.	Process and Processor Scheduling: Scheduling Criteria, First Come First Serve, Shortest Process Next, Round Robin, SRTN	
	Describe race condition & mutual exclusion, Semaphore, Monitors and Message Passing.	Inter-process Communication Principles of Concurrency, Mutual Exclusion, Software Approaches, Mutual Exclusion, Semaphores, Monitors Message Passing, Reader/Writer Problem	
	Identify Deadlocks and Apply Deadlock recovery procedure	Identification of Deadlocks (Deadlock conditions), Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, An Integrated Deadlock Strategy, Dining Philosophers Problem	
III	Describe memory management	Memory Management Memory Management requirements,	25%
	Differentiate Contiguous and Non-contiguous memory Differentiate physical and virtual primary memory	Contiguous Allocation: Partitioned memory allocation, Fixed & variable partitioning, Memory Allocation Strategies (First Fit, Best Fit, and Worst Fit), Replacement Policies (FIF LRU, Optimal, Other Strategies), Protection and Sharing Non contiguous allocation: Page allocation, Segmentation, Virtual Memory	
IV	Apply file management concepts in Operating System	File Management : File concept (File types, attributes and operation) ,	20%
	Explain Directory structure of Operating System	Directory Structure, Directory Implementation,	
	Explain File system implementation	File-System Implementation, Allocation Methods, Free-Space Management, File-System Mounting,	
	Implement file system security.	File Sharing and Protection	
V	Install Free & Open Source Software / Open	Unix commands and shell scripting	20%

	source Operating System	LINUX: installation and upgrade	
	Test and Execute basic Linux commands	Introduction to shell and commands: pwd, cd, mkdir, rmdir, ls, cat, cp, rm, mv, wc, split, differences (diff, cmp and comm), head, tail, grep, sort, who, uname, date, ps, apt-get install, apt-get remove	
	Test and Execute shell commands in a script	Editing files with “vi”, “vim”, “gedit”, “gcc” Linux Shell Basic shell scripts LINUX Shell Scripts	

Seminar:

- Emerging Operating System (Distributed, Parallel, Mobile, Cloud)
- Virtualization, VM ware,

3. SUGGESTED SPECIFICATION TABLE WITH HOURS & MARKS (THEORY)

Unit	Unit Title	Teaching Hours	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
I	Operating System Concepts	06	04	06	00	10
II	Processor & Process Management	11	06	10	04	20
III	Memory Management	09	06	08	02	16
IV	File Management	05	04	06	00	10
V	Linux Basics	08	02	04	08	14
Total		39	22	34	14	70

4. Text Books :

1. Operating Systems William Stallings, Pearson
2. Operating System Dhamdhare TMH
3. Operating Systems Concepts Galvin Silberschatz McGraw Hill
4. Modern Operating Systems Andrew S. Tanenbaum Pearson Edu. /PHI
5. Understanding Operating System Ida Flynn Brookes/Cole
6. Operating System P Balakrishna Prasad Scitech
7. Ashok Kumar Harnal, Linux –Application and administration, 2009 Edition, TMH
8. Sumitabha Das, Unix Concepts And Application, MGH

5. Webliography

1. Operating System concepts:
http://nptel.iitm.ac.in/courses/Webcoursecontents/IISc-ANG/Operating%20Systems/New_index1.html
2. Linux basics: www.freecss.com/guides/lsst
3. Linux basics: www.linuxcommand.org/writing_asell_scripts.php
4. Linux basics: www.distro.ibiblio.org/damnsmall/current/dsl-4.4.10-embedded.zip

Course Code	050020405			
Category	Software project			
Course Title	Advanced Web Application Development using PHP Framework Laravel			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	0	2	4	4
Pre-requisites (if any)	Knowledge of Programming Language and concepts of GUI			

Course Code	050020407			
Category	Software project			
Course Title	Web Application development using .Net Core			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	0	2	4	4
Pre-requisites (if any)	Knowledge of Programming Language and concepts of GUI			

General Guidelines for Software Project (Framework)

- Team Size: 3 Persons
 - It is suggested to develop Web/Desktop/Mobile applications using emerging frameworks like Magneto, CodeIgniter, Laravel, React.js, vue.js, Flutter, React Native, Angular, Node.js or any other open source frame works
- Project size in terms of Lines of Code (LOC) should be 1,000 lines or more(excluding comment lines). In case of extensive recursion used in medium to high complexity problem, this limit on LOC may be relaxed.
- Project definition should be such that it aims at providing solution to a specific requirement of real-life problem.
- The code should be self-documented and should follow coding standards (Style including (i) Generalized-Parameterized, (ii) Structured-Modular Coding Style, (iii) Compactness & Clarity, (iv) Checkpoints for intermediate results, (v) Naming Conventions, (vi) Self-Documented).
- Project must include
 - Atleast one listing/register type of report
 - Atleast 2 advanced reports

Expected Minimum Deliverables

- Project document and code must be uploaded on Github.
- Documentation of the project should include at least description of the problem, algorithm/pseudo code/flow chart (if any), functional testing and solution methodology.

- Data storage

Criteria for Evaluation of Software Projects

- Problem understanding and logic :
- Coding Standard :
- Completion and Operational :
- Reports

Course Code	050020406			
Category	MOOC			
Course Title	Open Elective (MOOC)			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	0	1	2	2
Pre-requisites (if any)				